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# Consumer-based brand equity measurement: lessons learned from an international study

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## Abstract

**Purpose** – The purpose of this paper is to examine the performance of Aaker's dominant conceptualization of consumer-based brand equity (brand awareness, brand associations, perceived quality and brand loyalty) in a multi-national and multi-sector European context and highlights important lessons *vis-à-vis* the measurement of brand assets across countries.

**Design/methodology/approach** – Cross-category data was collected through a survey over a period of two months from a representative sample of consumers in three European countries ($n = 1,829$), the UK ($n = 605$), Germany ($n = 600$) and Greece ($n = 624$).

**Findings** – The findings suggest that Aaker's dimensions of consumer-based brand equity cannot be clearly separated. More specifically the dimensions of brand awareness, brand associations and brand loyalty could not be always clearly discriminated in all national contexts.

**Originality/value** – The paper contributes to the limited amount of cross-national research on brand equity by assessing the most widely used conceptualization of consumer-based brand equity. Contrary to previous research, this study has used data from real consumers who evaluated a range of brands across product categories (including goods, services and internet).

**Keywords** Loyalty, Cross-national, Perceived quality, Awareness, Aaker, Associations, Consumer-based brand equity

**Paper type** Research paper

## Introduction

Brand equity is a key concept for marketing academics and one of the most prized assets for firms (Ambler, 2003; Christodoulides and de Chernatony, 2010). According to Interbrand (2013), Apple is presently the most valuable brand in the world with its value estimated at 98.3 billion dollars. This market-based intangible asset reflects the value of the brand for different stakeholders the main of which have been identified as the consumer and the firm. Marketing practitioners and academics alike regard brand equity as a platform upon which to build a competitive advantage, secure future cash flows and grow shareholder wealth (Keller, 2008; Kerin and Sethuraman, 1998).

Brands are no more "unique" than various other intangible assets and although they are often treated differently from tangible assets they, in fact, possess characteristics that allow them to be managed similar to other tangible and intangible assets (El-Tawy and Tollington, 2008). It is, therefore, important to define brand equity, to identify its

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constituent dimensions and to assess the suitability of various conceptualizations and measurements in multi-country environments.

The conceptualization and measurement of brand equity, including its sources and outcomes, are challenging tasks, particularly in an international context (Kish *et al.*, 2001). In principle, there is no agreement on the dimensionality of brand equity. In their literature review, Christodoulides and de Chernatony (2010) have identified several dimensions of consumer-based brand equity used in previous research. Similarly, practitioners suggest a number of aspects that could be considered dimensions of consumer-based brand equity (Veloutsou *et al.*, 2013). Despite the multiplicity of voices with regards to the composition of brand equity, it appears that the most commonly adopted model of consumer-based brand equity's conceptual domain in empirical studies is Aaker's (1991). He identified brand awareness, brand associations, perceived quality and brand loyalty as the consumer-based components of brand equity but has never operationalized his model. The lack of generally accepted scales to measure the brand assets means that it is more difficult for practitioners to justify investment in brand building initiatives.

The majority of the published research on consumer-based brand equity has focused primarily on a single country – namely the USA (see Christodoulides and de Chernatony, 2010; Lehmann *et al.*, 2008). With the increasing globalization of markets and brands it becomes more important than ever to adopt measures of brand equity that can be applied in an international context allowing valid comparisons between domestic and foreign markets. The conditions within which brands are competing in various contexts may vary greatly from country to country and from category to category. Valid measures of brand equity that work across foreign markets and product categories would allow practitioners to more accurately inform their brand positioning and communications strategies and researchers to advance knowledge of international branding. Research on consumer-based brand equity measurement in an international context remains extremely limited, focused on goods brands and reliant on undergraduate student samples (e.g. Yoo and Donthu, 2001). Given the limitations of the previous research, it is not surprising that researchers are calling for substantially more research in varying product categories (Whitelock and Fastoso, 2007) including service and internet brands that prominently feature in the "Top 100 Global Brands" ranking published annually by Interbrand (2013).

This paper contributes to the scarce research on international brand equity measurement by looking into the performance of a scale around Aaker's dominant conceptualization of consumer-based brand equity in a European context through data from three European countries, the UK, Germany and Greece. Whilst previous research (e.g. Buil *et al.*, 2008; Yoo and Donthu, 2001) has attempted to examine a scale of consumer-based brand equity using the Aaker's conceptualization and using cross-national data, the sample used in some occasions comprised undergraduate students rather than real consumers whilst the brands assessed were pre-specified and from a limited number of product categories. This paper provides an assessment of consumer-based brand equity using data from a varied sample of real consumers in three European countries involving a range of different brands from six product categories spanning goods, service and internet domains.

The paper opens by reviewing the conceptualization of consumer-based brand equity and the limited research on brand equity in an international context. It then goes on to present the methodology adopted for collecting data around Aaker's dimensions of consumer-based brand equity from three European countries. The analysis reveals issues with the applicability of Aaker's conceptualization of consumer-based brand equity in a cross-national setting as well as with the discriminant validity of its constituent dimensions. The paper concludes with a discussion of the key lessons learned, and a call to researchers to revisit the consumer brand equity construct, re-evaluate how its measurement is approached, and to think more creatively about how consumer-based brand equity can be integrated into international marketing theories in order to provide richer insights for marketing practice.

## Conceptualization of consumer-based brand equity

Brand equity is a key intangible asset that arises from past brand building activities and encompasses the added value endowed by the brand to the product (Farquhar, 1989). Research has, hitherto, established a link between brand equity and desired business outcomes including a brand's extension potential (e.g. Aaker and Keller, 1990), WOM recommendation (e.g. Vázquez *et al.*, 2002), price premium (e.g. Ailawadi *et al.*, 2003), attitudes (e.g. Yoo and Donthu, 2001) and purchase intentions (e.g. Yoo and Donthu, 2001). One of the first and most widely used definitions of brand equity is from Aaker (1991), who defines it as a "a set of assets and liabilities linked to a brand, its name and symbol, that add to or subtract from the value provided by a product or service to a firm and/or that firm's customers" (p. 15). This definition suggests two main recipients of brand value based on which two streams of research have been developed: consumer-based and firm-based brand equity. Whilst the majority of research on brand equity has approached the brand asset from either the firm or the consumer perspective some researchers have also examined the value of the brand for other internal (such as employees – see, e.g. King and Grace, 2010) as well as external stakeholder groups (such as channel members – see, e.g. Nyadzayo *et al.*, 2011).

Firm-based brand equity is primarily concerned with placing a financial value to the brand asset (Simon and Sullivan, 1993) and is often perceived as an outcome of consumer-based brand equity which in turn focuses on the (additional) value that the consumer derives from the brand name (Farquhar, 1989). Consumer-based brand equity elaborates largely on the idea that the strength of a brand lies in the minds of consumers (Leone *et al.*, 2006). Keller (1993, p. 8) draws on cognitive psychology to define consumer-based brand equity as the "differential effect of brand knowledge on consumer response to the marketing of the brand". The development of consumer-based brand equity thus necessitates consumers to be aware of the brand name and moreover to hold strong, favourable and unique associations of the brand in memory. Similar to Aaker (1991, 1996), Keller has never operationalized the two dimensions of consumer-based brand equity.

Although a substantial body of research on brand equity exists, there is little consensus on what brand equity is and what its constituent dimensions are (Christodoulides and de Chernatony, 2010; Maio Mackay, 2001; Veloutsou *et al.*, 2013). However, the majority of researchers agree that the concept is multi-dimensional whilst the divergence of views as to its conceptualization may arguably be the product of the three blind men and elephant syndrome (Ambler, 2003; Christodoulides and de Chernatony, 2010).

Based on their literature review, Christodoulides and de Chernatony (2010) identified various dimensions of consumer-based brand equity used by researchers including but not limited to social image (Lassar *et al.*, 1995), performance (Lassar *et al.*, 1995), trust (Burmann *et al.*, 2009; Christodoulides *et al.*, 2006), relationships (Blackston, 1992; Sharp, 1995), awareness (Aaker, 1991; Berry, 2000; Im *et al.*, 2012; Keller, 1993), perceived quality (Aaker, 1991; Kamakura and Russell, 1993; Netemeyer *et al.*, 2004; Pappu *et al.*, 2005; Yoo and Donthu, 2001), loyalty (Aaker, 1991; de Chernatony *et al.*, 2004; Im *et al.*, 2012; Pappu *et al.*, 2005; Washburn and Plank, 2002; Yoo and Donthu, 2001), image (Im *et al.*, 2012; Keller, 1993; Sharp, 1995), associations (Aaker, 1991; Im *et al.*, 2012; Keller, 1993; Netemeyer *et al.*, 2004; Pappu *et al.*, 2005; Yoo and Donthu, 2001), brand meaning (Berry, 2000), brand benefit uniqueness/clarity (Burmann *et al.*, 2009; Netemeyer *et al.*, 2004), satisfaction (Blackston, 1992; de Chernatony *et al.*, 2004; Ha *et al.*, 2010), reputation (de Chernatony *et al.*, 2004), willingness to pay a premium (Netemeyer *et al.*, 2004), value (Shankar *et al.*, 2008) and symbolic and functional utility (Kocak *et al.*, 2007; Vázquez *et al.*, 2002).

Although there are several conceptualisations, Aaker's (1991) dimensionality of consumer-based brand equity is indeed the most commonly used in empirical research. Aaker (1991, 1996) identifies the dimensions of consumer-based brand equity as brand awareness, brand associations, perceived quality and brand loyalty. Brand awareness is defined as the "the ability of a potential buyer to recognize or recall that a brand is a member of a certain product category (Aaker, 1991, p. 61). Brand associations refer to "anything linked in memory to a brand" (Aaker, 1991, p. 109). Perceived quality is "the consumer's judgment about a product's overall excellence or superiority" (Zeithaml, 1988, p. 3) whilst brand loyalty is "the attachment that a customer has to a brand" (Aaker, 1991, p. 39). Aaker himself has never operationalized a scale to measure brand equity but other researchers did (see, e.g. Atilgan *et al.*, 2005; Bravo *et al.*, 2007; Buil *et al.*, 2008; Jung and Sung, 2008; Kim and Hyun, 2011; Konecnik and Gartner, 2007; Lee and Back, 2010; Pappu *et al.*, 2005; Pike *et al.*, 2010; Tong and Hawley, 2009; Washburn and Plank, 2002; Yasin *et al.*, 2007; Yoo *et al.*, 2000; Yoo and Donthu, 2001).

Aaker's aforementioned dimensionality of consumer-based brand equity has been applied in several contexts and problems have been reported with regard to the discriminant validity of this conceptualisation, particularly in so far as awareness fails to discriminate with associations (e.g. Kim and Hyun, 2011; Washburn and Plank, 2002; Yoo and Donthu, 2001). Conceptually some researchers also regard brand loyalty as an outcome rather than a dimension of brand equity (e.g. Chaudhuri and Holbrook, 2001). In order to be able to make recommendations to managers on how to manage their brand equity, or study the nomological network of its constituent components, we need to generate a better understanding of the composition of brand equity in disparate cultural contexts and distinct product categories.

## International brand equity: the research gap

Companies working in international markets have been trying for years to find ways to measure the equity of their brands across markets in a comparable manner (see Kish *et al.*, 2001; Wright and Nancarrow, 1999). The challenges for companies that not only operate across multiple countries but also across multiple sectors are even greater (Kish *et al.*, 2001). Similar challenges are faced by researchers who wish to test marketing theories containing brand equity in cross-national, multi-sector settings.

Empirical research on brand equity has focused largely on single country data, particularly from the USA, resulting from evaluations of brands in goods domains. With the increasing globalization of markets and the need to develop strong global brands to compete in international environments (Hsieh, 2004), it is indeed surprising that only little research exists on the role of (consumer-based) brand equity in international marketing using consumer data, summarized in Table I.

**Table I.** Studies of brand equity with international samples

| Paper | Countries | Product categories | Conceptualization of brand equity | Sample | Sample size |
|---|---|---|---|---|---|
| Yoo and Donthu (2001) | USA and South Korea | Preselected brands of: athletic shoes, film and colour TV sets | Aaker's dimensions resulting to:<br>Brand Loyalty<br>Perceived Quality<br>Awareness/associations | Undergraduate students | USA – 196<br>South Korea - 218 |
| Hsieh (2004) | Australia, Belgium, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, The Netherlands, Russia, South Korea, Spain, Taiwan, Thailand, Turkey, UK and USA | Preselected brands of: Automobiles | Data from a data set owned by MORPACE International, a multi-national research firm with measures for:<br>Brand recognition<br>Brand attachment<br>Market size | General public living in various cities in each country | From 44 to 189 in each country, a total of 2,828 respondents |
| Buil *et al.* (2008) | UK and Spain | Preselected brands of: soft drinks, sportswear, cars and consumer electronics | Aaker's dimensions and in particular:<br>Brand awareness<br>Perceived quality<br>Brand loyalty<br>Brand associations: perceived value<br>Brand associations: brand personality<br>Brand associations: organization | Birmingham (UK)<br>Zaragoza (Spain) quota sampling | UK – 411<br>Spain - 411 |
| Jung and Sung (2008) | US and South Korea | Three preselected brands (Polo, Gap and Levi's) | Used the multi-dimensional brand equity (MBE) and overall brand equity (OBE) models developed by Yoo and Donthu (2001) | College Students | Americans in the USA 100<br>South Koreans in the USA 100<br>South Koreans in South Korea 100 |
| Lehmann *et al.* (2008) Study 1 | USA and China | Preselected brands of: Soft drinks | 27 dimensions of brand performance were suggested. They were generated from the literature, including Aaker (1996), Fournier (1998), Ambler (2003), Keller (2002, 2008) and Keller and | Chicago (USA)<br>Shanghai (China) | USA - 100<br>China – 100 |

(*continued*)

| Paper | Countries | Product categories | Conceptualization of brand equity | Sample | Sample size |
|---|---|---|---|---|---|
| | | | Lehmann (2003) and reports from commercial brand tracking approaches, including Young and Rubicam's Brand Asset Valuator (BAV), Millward Brown and Research International | | |
| Lehmann *et al.* (2008) Study 2 | USA and China | Preselected brands of: Soft drinks Toothpaste Fast Food | The same dimensions of Study 1, measured with 3 items per dimension | Chicago (USA) Shanghai (China) | USA -150 China – 150 |
| Broyles *et al.* (2010) | USA and China | Preselected brand: KFC | Functional aspect consisting from perception of a brand's performance and quality (perceived performance and perceived quality) Experiential component consisting from brand's resonance and imagery | University students | USA – 278 China – 300 |

**Table I.**

Following an etic approach (Berry, 1969), Yoo and Donthu (2001) purported to develop a scale of consumer-based brand equity around Aaker's four dimensions using data from a sample of American, Korean-American and Korean students. The resultant scale, which was developed to be invariant across the three samples, failed to discriminate between awareness and associations – two theoretically distinct concepts. When the study was replicated with three samples of college students (Americans in the USA, South Koreans in the USA and South Koreans in South Korea), some differences in the importance of the various dimensions of brand equity were reported (Jung and Sung, 2008). Also, the application of this scale in both studies was limited to goods brands (athletic shoes, television sets and films in the first study and clothes brands in the second study).

Buil *et al.* (2008) followed the procedure of Steenkamp and Baumgartner (1998) to establish the invariance of a scale around Aaker's consumer-based brand equity dimensions using data from real consumers in the UK and Spain. However, similar to the study of Yoo and Donthu (2001), only goods brands were used in this research (soft drinks, sportswear, consumer electronics and cars) rendering the scale's applicability in other domains questionable. Other cross-national research on brand equity has focused on models to decompose brand equity (e.g. Hsieh, 2004), the functional and the experiential component of brand equity (Broyles *et al.*, 2010) or on dimensions of brand metrics (Lehmann *et al.*, 2008) with an emphasis again on goods categories (e.g. automobiles, soft drinks and toothpaste) and comparative data from the USA and other countries such as China.

In summary, only a few studies have examined (consumer-based) brand equity across countries. All of the existing cross-national research of brand equity that has as a starting point the conceptualisation of the construct has data collected from two national contexts that appear to be selected on the basis of convenience, while pre-specified brands were given to the respondents. Furthermore, there is no evidence available on the validity and/or invariance of the scales used beyond goods brands. However, service and internet brands such as Google, McDonalds, Disney, HSBC and Amazon are consistently recognized amongst the world's most valuable brands (Interbrand, 2013). This paper focuses on testing the measurement invariance of the most dominant conceptualisation of consumer-based brand equity (i.e. Aaker's 1991, 1996) by collecting consumer data from three European countries on a range of brands including goods, services and internet brands.

## Methodology

### *Instrument development*

As intimated earlier, despite defining brand equity and specifying its constituent consumer-based dimensions as brand awareness, associations, perceived quality and brand loyalty, Aaker (1991, 1996) never went on to develop a measure for this. Although other researchers have developed and validated scales around Aaker's dimensions (e.g. Buil *et al.*, 2008; Pappu *et al.*, 2005; Yoo and Donthu, 2001), none was free of problems. For example, the Yoo and Donthu (2001) scale collapses brand awareness with brand associations whilst Pappu *et al.* (2005) use a single item measure for brand awareness. Thus, rather than taking a single existing measure of brand equity, we bring together measures from multiple studies to expand the item pool and measure Aaker's notions of brand awareness, associations, perceived quality and brand loyalty. These were turned into a mini questionnaire and were sent to a panel of experts to be judged for face validity (in line with the recommended procedures of

Hardesty and Bearden, 2004). Eight experts (senior academics with significant publications in the area of brand equity) responded to our invitation and rated the items based on clarity and representativeness of the content domain. Out of 35 items in the initial pool (six for awareness, seven for associations, 11 for perceived quality and 11 for loyalty – Appendix), 12 items survived this process (three for each one of the dimensions) and were subsequently included in a survey questionnaire in keeping with previous research on consumer-based brand equity. All items were measured on seven-point scales. The questionnaire was first pre-tested in the UK using a sample of students. Following that, the instrument was "back translated" into German and Greek by marketing academics native speakers of German and Greek with fluency in English. Two academics with this profile were involved in each country; the first one translated the instrument into the required language while the second one back translated it into English. The two back translated versions were then passed on to a marketing academic who is a native speaker of English to check if they were the same with the original English version and to confirm they have semantic equivalence (Mallinckrodt and Wang, 2004). Where there were queries on the meaning, the Greek and German questionnaires were adjusted. The survey was pre-tested in German and in Greek and then it was administered through face-to-face interviews by trained fieldworkers predominantly in urban centres.

*Sampling and data collection*

Quotas were set for age and gender based on census data from each country to ensure the samples were collected from multiple areas per country representative of national populations. The fieldworkers were asked to recruit respondents who had been residents in each one of the countries. Data were collected from various locations in each country. Each respondent was asked to pick a category (from a list of categories that included goods – e.g. shampoo; service – e.g. coffee shop, bank; and internet – e.g. retailer) and to identify their favourite (or least favourite) brand from that category. The dilemma we faced was that whilst we wanted our respondents to have knowledge of the brands they would evaluate, at the same time we wanted to avoid having respondents only pick their preferred or currently used brand as this would predominantly capture high equity brands. On the other hand a pre-selection of specific brands from the identified categories would pose problems of equivalence across the countries especially for low equity brands (which may, indeed, be smaller local brands). By asking the respondents to respond to the questions in relation to their most favourite or least favourite brand ensured that: a large number of different brands was evaluated (rather than a small number of pre-specified brands) and that the data was not severely skewed towards high equity brands. Consumers were asked to respond to subsequent questions with regard to their chosen brand. The data collection took place simultaneously in the three European countries (over a period of two weeks) and produced a total of 1,829 fully completed questionnaires as reported in Table II (600 in Germany, 605 in UK and 624 in Greece and 960 for good brands, 736 for services brands and 132 for internet brands). As an incentive to participate, survey respondents were offered the chance to win one of two gift cards (in each country) worth £100.

## Results

It is highly desirable for any scale to possess high levels of measurement invariance, in order to allow findings using the measures to be compared across nations. This is

| | Germany | | UK | | Greece | |
|---|---|---|---|---|---|---|
| | *n* | % | *n* | % | *n* | % |
| Gender | | | | | | |
| Male | 305 | 50.41 | 289 | 48.17 | 311 | 49.84 |
| Female | 300 | 49.59 | 310 | 51.67 | 312 | 50.00 |
| na | 0 | 0.00 | 1 | 0.17 | 1 | 0.16 |
| Age | | | | | | |
| 18-24 | 78 | 12.89 | 81 | 13.50 | 91 | 14.58 |
| 25-44 | 213 | 35.21 | 190 | 31.67 | 226 | 36.22 |
| 45-64 | 200 | 33.06 | 190 | 31.67 | 184 | 29.49 |
| 65+ | 114 | 18.84 | 139 | 23.17 | 122 | 19.55 |
| na | 0 | 0.00 | 0 | 0.00 | 1 | 0.16 |
| Years in the country | | | | | | |
| All my life in | 560 | 92.56 | 502 | 83.67 | 551 | 88.30 |
| More than 10 | 2 | 0.33 | 71 | 11.83 | 61 | 9.78 |
| 5-10 | 33 | 5.45 | 19 | 3.17 | 6 | 0.96 |
| Less than 5 years | 10 | 1.65 | 8 | 1.33 | 5 | 0.80 |
| na | 0 | 0.00 | 0 | 0.00 | 1 | 0.16 |
| Education | | | | | | |
| Higher degree/postgraduate | 78 | 12.89 | 21 | 3.50 | 53 | 8.49 |
| Degree, or degree equivalent | 218 | 36.03 | 152 | 25.33 | 236 | 37.82 |
| School leaving certificate | 151 | 24.96 | 345 | 57.50 | 211 | 33.81 |
| Other | 111 | 18.35 | 79 | 13.17 | 116 | 18.59 |
| No qualification | 37 | 6.12 | 3 | 0.50 | 8 | 1.28 |
| Employment | | | | | | |
| In full-time employment | 259 | 42.81 | 239 | 39.83 | 284 | 45.51 |
| In part-time employment | 72 | 11.90 | 88 | 14.67 | 79 | 12.66 |
| Retired | 111 | 18.35 | 137 | 22.83 | 131 | 20.99 |
| Unemployed | 23 | 3.80 | 12 | 2.00 | 60 | 9.62 |
| Other | 126 | 20.83 | 124 | 20.67 | 70 | 11.22 |
| na | 1 | 0.17 | 0 | 0.00 | 0 | 0.00 |
| Income | | | | | | |
| Lower than €10,000 | 0 | 0.00 | 141 | 23.50 | 0 | 0.00 |
| €10,000-€20,000 | 49 | 8.10 | 139 | 23.17 | 0 | 0.00 |
| €20,001-€30,000 | 75 | 12.40 | 91 | 15.17 | 109 | 17.47 |
| €30,001-€40,000 | 47 | 7.77 | 76 | 12.67 | 39 | 6.25 |
| €40,001-€50,000 | 44 | 7.27 | 78 | 13.00 | 21 | 3.37 |
| €50,001-€60,000 | 32 | 5.29 | 49 | 8.17 | 8 | 1.28 |
| €60,001-€70,000 | 9 | 1.49 | 24 | 4.00 | 3 | 0.48 |
| €70,000+ | 131 | 21.65 | 2 | 0.33 | 13 | 2.08 |
| na | 141 | 23.31 | 0 | 0.00 | 5 | 0.80 |
| Product category | | | | | | |
| Goods | 332 | 55.33 | 326 | 53.88 | 302 | 48.40 |
| Services | 214 | 35.67 | 236 | 39.01 | 286 | 45.83 |
| Internet brands | 54 | 9.00 | 42 | 6.94 | 36 | 5.77 |
| Total | 605 | 100.00 | 600 | 100.00 | 624 | 100.00 |

**Table II.**
Sample profile

particularly true in the case of brand equity measures given the international nature of markets that most brands operate in. Accordingly, the items were assessed for validity via confirmatory factor analysis (CFA) using LISREL 8.50's multi-group function. In line with procedures outlined in the literature (Steenkamp and Baumgartner, 1998; Vandenberg and Lance, 2000) for measurement assessment, an omnibus test was undertaken. The omnibus test imposes the strictest level of invariance necessary (as required by the uses the measures will be put to).

In the case of the current measures, we decided to test for a strong level of invariance, and so the omnibus test simultaneously assessed the measures across the three countries for configural, metric, scalar and factor variance invariance. On running this omnibus test, $\chi^2$ was significant ($\chi^2 = 1{,}476.93$; df = 192; $p < 0.001$), indicating that the measurement model does not exactly mirror the data. Moreover, relative $\chi^2$ ($\Delta\chi^2/\Delta$df = 7.69) was greater than the acceptable range of two to three as recommended by Carmines and McIver (1981). The approximate fit heuristics indicated mixed results. Although RMSEA was higher than the recommended value of 0.08, the CFI, IFI and NNFI were within reasonable levels (all were 0.93) and as such, this indicates that the items we recovered from the literature to include as measures of Aaker's measurement structure, under the battery of strict invariance constraints across the three countries, do what might be considered only a moderately good job in describing the data (CFI = 0.93, IFI = 0.93, NNFI = 0.93 RMSEA = 0.105) (Bentler, 1990; Browne and Cudeck, 1993). Table III shows the factor loadings for each of the three samples. All loadings were greater than the recommended cut-off value of 0.5 and were statistically significant (Hair *et al.*, 1998). Composite reliability estimates for each sub-scale of consumer-based brand equity were higher than the recommended 0.70 value (Bagozzi and Yi, 1988) indicating adequate levels of internal consistency. Likewise, AVEs were greater than the 0.50 cutoff point supporting the convergent validity of consumer-based brand equity's dimensions. However, further examination into the measures reveals that Aaker's dimensions have discrimination problems with each other in all three countries (AVEs lower than the squared correlations, see Fornell and Larcker, 1981).

Table IV shows lack of discriminant validity for brand awareness and brand associations in Germany and the UK and also for brand associations, brand loyalty and perceived quality in Germany and in Greece. A closer examination into the lack of discriminant validity led us to repeat this analysis separately for different types of brands, i.e. goods/services and internet. Table IV suggest that the issues of discriminant validity amongst dimensions of consumer-based brand equity are more severe for service and internet brands (Table V).

**Table III.** Measurement analysis results – parameter estimates

| Items | Germany $\lambda$ | UK $\lambda$ | Greece $\lambda$ | Common $\lambda$ | *t*-value |
|---|---|---|---|---|---|
| I have heard of this brand | 0.637 | 0.704 | 0.671 | 0.669 | 28.71 |
| I am quite familiar with this brand | 0.861 | 0.835 | 0.890 | 0.861 | – |
| I can recognize this brand among other brands | 0.762 | 0.725 | 0.743 | 0.743 | 31.83 |
| This brand has strong associations | 0.723 | 0.845 | 0.731 | 0.760 | 25.04 |
| This brand has favourable associations | 0.846 | 0.869 | 0.875 | 0.863 | 26.46 |
| It is clear what this brand stands for | 0.584 | 0.617 | 0.592 | 0.597 | – |
| This brand is good quality | 0.938 | 0.948 | 0.948 | 0.945 | 78.49 |
| This brand has excellent features | 0.945 | 0.959 | 0.613 | 0.782 | 70.96 |
| Compared to other brands in its category, this brand is of very high quality | 0.940 | 0.910 | 0.953 | 0.934 | – |
| I feel loyal to this brand | 0.887 | 0.912 | 0.944 | 0.914 | 72.32 |
| This brand is my first choice | 0.915 | 0.944 | 0.946 | 0.935 | – |
| I am committed to this brand | 0.918 | 0.938 | 0.952 | 0.936 | 77.08 |

Table IV.
Aaker's brand equity model: correlations (squared correlation), CR & AVE

| | AWR | ASS | LOY | QLT |
|---|---|---|---|---|
| Germany | | | | |
| AWR | *0.58* | | | |
| ASS | 0.77(0.59) | *0.53* | | |
| LOY | 0.47(0.22) | 0.73(0.53) | *0.82* | |
| QLT | 0.48(0.23) | 0.76(0.57) | 0.84(0.70) | *0.89* |
| CR | 0.80 | 0.77 | 0.93 | 0.96 |
| UK | | | | |
| AWR | *0.57* | | | |
| ASS | 0.77(0.59) | *0.62* | | |
| LOY | 0.35(0.12) | 0.40(0.16) | *0.87* | |
| QLT | 0.37(0.14) | 0.51(0.26) | 0.75(0.56) | *0.88* |
| CR | 0.80 | 0.83 | 0.95 | 0.96 |
| Greece | | | | |
| AWR | *0.60* | | | |
| ASS | 0.67(0.45) | *0.55* | | |
| LOY | 0.54(0.30) | 0.78(0.60) | *0.90* | |
| QLT | 0.57(0.32) | 0.79(63) | 0.87(0.75) | *0.73* |
| CR | 0.82 | 0.78 | 0.96 | 0.88 |

**Notes:** AWR, Awareness; ASS, associations; LOY, loyalty; QLT, quality; CR, composite reliability. The italised in the diagonal shows AVE

## Discussion

The reliability and validity of constructs as well as the equivalence of respective measures across national borders are key considerations in international marketing research (e.g. Buil *et al.*, 2008; Cadogan *et al.*, 1999; Steenkamp and Baumgartner, 1998). In this paper we examine the most widely used dimensionality to measure consumer-based brand equity, that is Aaker's (1991, 1996), using data from a sample of real consumers in three European countries (Germany, UK and Greece) who evaluated a range of goods, service and internet brands.

Our results reveal that in the three countries under investigation the measurement invariant model that was capturing Aaker's dimensions of brand equity did not exactly mirror the data. Even within a seemingly homogenous geographical cluster, such as Europe, the results from the invariance analysis suggested differences in the ways consumers perceived and evaluated brand equity in Germany, UK and Greece. This contradicts previous research from Yoo and Donthu (2001) and Buil *et al.* (2008) who found consumer-based brand equity to be equivalent across the USA and Korea, and the UK and Spain, respectively. An explanation for this may be the inclusion and evaluation of a more diverse set of brands in the present study (including service and internet brands) whilst the aforementioned studies focused exclusively on goods brands. This may suggest that consumer-based brand equity for goods brands is more uniform, but for more intangible and abstract service/internet brands (de Chernatony *et al.*, 2011), variations between nations are likely to occur. Another possible explanation may be due to the socio-economic differences amongst the three countries amidst the European debt crisis, which has differentially affected consumers' spending power in the countries under investigation. The financial crisis in Europe has inevitably hit Greece more than Germany and the UK, and arguably the rubric of relationships between consumers and brands in Greece may have

| | AWR | ASS | LOY | QLT |
|---|---|---|---|---|
| (a) Goods brands | | | | |
| Germany | | | | |
| AWR | *0.57* | | | |
| ASS | 0.74(0.55) | *0.61* | | |
| LOY | 0.49(0.24) | 0.66(0.44) | *0.80* | |
| QLT | 0.54(0.29) | 0.69(0.48) | 0.82(0.67) | *0.90* |
| CR | 0.80 | 0.82 | 0.92 | 0.96 |
| UK | | | | |
| AWR | *0.52* | | | |
| ASS | 0.78(0.61) | *0.70* | | |
| LOY | 0.45(0.20) | 0.37(0.14) | *0.88* | |
| QLT | 0.48(0.23) | 0.50(0.25) | 0.68(0.46) | *0.86* |
| CR | 0.76 | 0.87 | 0.96 | 0.95 |
| Greece | | | | |
| AWR | *0.58* | | | |
| ASS | 0.67(0.45) | *0.62* | | |
| LOY | 0.63(0.40) | 0.81(0.66) | *0.91* | |
| QLT | 0.67(0.45) | 0.85(0.72) | 0.89(0.79) | *0.67* |
| CR | 0.80 | 0.83 | 0.97 | 0.85 |
| (b) Service brands | | | | |
| Germany | | | | |
| AWR | *0.59* | | | |
| ASS | 0.76(0.58) | *0.47* | | |
| LOY | 0.44(0.19) | 0.80(0.64) | *0.87* | |
| QLT | 0.43(0.18) | 0.83(0.69) | 0.86(0.74) | *0.84* |
| CR | 0.81 | 0.72 | 0.94 | 0.95 |
| UK | | | | |
| AWR | *0.62* | | | |
| ASS | 0.76(0.58) | *0.53* | | |
| LOY | 0.31(0.10) | 0.45(0.20) | *0.96* | |
| QLT | 0.36(0.13) | 0.57(0.32) | 0.79(0.62) | *0.97* |
| CR | 0.83 | 0.79 | 0.96 | 0.97 |
| Greece | | | | |
| AWR | *0.61* | | | |
| ASS | 0.67(0.45) | *0.50* | | |
| LOY | 0.47(0.22) | 0.74(0.55) | *0.89* | |
| QLT | 0.50(0.25) | 0.74(0.55) | 0.85(0.72) | *0.92* |
| CR | 0.82 | 0.74 | 0.96 | 0.97 |
| (c) Internet brands | | | | |
| Germany | | | | |
| AWR | *0.56* | | | |
| ASS | 0.72(0.52) | *0.47* | | |
| LOY | 0.32(0.10) | 0.76(0.58) | *0.85* | |
| QLT | 0.44(0.19) | 0.87(0.76) | 0.89(0.79) | *0.91* |
| CR | 0.79 | 0.71 | 0.94 | 0.97 |
| UK | | | | |
| AWR | *0.64* | | | |
| ASS | 0.82(0.67) | *0.57* | | |

(*continued*)

**Table V.**
Correlations (squared correlation), CR and AVE

| | AWR | ASS | LOY | QLT |
|---|---|---|---|---|
| LOY | 0.31(0.10) | 0.32(0.10) | *0.70* | |
| QLT | 0.34(0.12) | 0.39(0.15) | 0.90(0.81) | *0.89* |
| CR | 0.84 | 0.80 | 0.87 | 0.95 |
| Greece | | | | |
| AWR | *0.61* | | | |
| ASS | 0.80(0.64) | *0.50* | | |
| LOY | 0.65(0.42) | 0.77(0.59) | *0.91* | |
| QLT | 0.60(0.36) | 0.71(0.50) | 0.78(0.61) | *0.92* |
| CR | 0.82 | 0.75 | 0.97 | 0.97 |

**Notes:** AWR, Awareness; ASS, associations; LOY, loyalty; QLT, quality. The italized in the diagonal shows AVE CR, composite reliability

**Table V.**

changed or indeed be different from the other two countries. Many brands in Greece were forced to significantly decrease their advertising budgets (Edgecliffe-Johnson, 2012) which based on previous research may reduce levels of awareness and strength of associations (see, e.g. Yoo *et al.*, 2000).

Furthermore, issues emerged with regards to the discriminant validity of the constituent dimensions of consumer-based brand equity. More specifically, brand awareness and brand associations failed to discriminate in Germany and the UK, whilst the discriminant validity of brand associations with perceived quality and brand loyalty also proved problematic in both Germany and in Greece. These results are consistent with previous research reporting problems with the discriminant validity of Aaker's model across different country samples (e.g. Yoo and Donthu, 2001) or indeed with single country data (Kim and Hyun, 2011). For instance, brand awareness and brand associations, two theoretically distinct constructs which jointly form brand knowledge (Keller, 1993) were previously collapsed into one dimension by Yoo and Donthu (2001) due to lack of discriminant validity. Therefore, one cannot argue with confidence that the items used to measure these consumer-based brand equity dimensions are doing an adequate job in assessing the constructs. In fact, the dimensions of awareness and associations are conceptually similar, and the results of the current study, taken together with previous efforts to measure these dimensions provides food for thought when it comes to approaching the assessment of consumer-based brand equity dimensions. Specifically, following conventional practice within the academic community (Aaker, 1996; Coleman *et al.*, 2011), we utilized an approach to assess the brand equity dimensions that asked respondents to focus on brand associations at a generic level, in this case in terms of strength/favorability (see Keller, 1993) to allow meaningful comparisons across brands in various product categories. For instance, we used items from the literature that asked respondents how much they agreed with the statement that it is "clear what this brand stands for". We also asked respondents standard questions to assess awareness, such as "I have heard of this brand". On reflection, it seems that Awareness is a condition that consumers need to possess in order for them to be able to also have associations in terms of understanding what the brand stands for. As a result, it seems unreasonable for items that measure Awareness to not cross-load with items that also measure Associations, and so on reflection, it is likely that there is a good chance that there will always be some discrimination problems with Awareness and Associations measures.

A solution to this problem may be to make the measurement less generic, and more specific to certain brands. For instance, by forcing respondents to consider a specific brand, and consider highly brand-specific association information (e.g. the brand's positioning strategy or product characteristics), we may be able to capture awareness at the abstract level, and associations at a more tangible level. In so doing, one may be able to discriminate more easily the two constructs. Whilst this might potentially overcome problems of discriminant validity, it would at the same time hinder the applicability of the scale across brands and product categories, and thus comparability. This is a particular challenge for researchers who often use a wide range of brands from various market contexts to test their theories.

Furthermore, there is no guarantee that such an approach would overcome the conditionality effect completely, since awareness is a precondition for associations. As a result, even by modifying the item wording strategy (i.e. by capturing a Associations at a very specific brand level), problems with discriminant validity may still emerge.

The way in which brand equity is built and nurtured might differ from country to country and from context to context, due to various differences in the micro and macro environment. The measurement of certain characteristics of brands, such as brand personality, may need to be adjusted in different cultural environments so as to be valid (see Aaker *et al.*, 2001; Rojas-Mendez *et al.*, 2004). Contrary to the desire of international marketers to apply a consumer-based brand equity scale across markets this study corroborates previous research suggesting that brand equity may indeed be culture specific (e.g. Kocak *et al.*, 2007). This finding raises concerns about the discretionary application of the consumer-based brand equity construct and its measures from one context to another without an assessment of equivalence.

Alternatively, a more statistical solution would be to statistically orthogonalize the Awareness and Associations measures following Cadogan *et al.* (2001). Specifically, one could partial out the variance in the Associations items that is shared with the Awareness items – the resulting Associations scores could be used as purified measures of Associations that are free of Awareness information. The partialing out of variance may work at a statistical level, but it is a cumbersome approach, and the resulting numerical values obtained for the Associations scores would be hard to understand in an intuitive sense (since they are essentially transformations of the original data). As a result, it becomes hard to use the scores for practical purposes, or to interpret their meaning in substantive tests of theory.

In the following section, we reflect further on the issue of measuring brand equity using Aaker's conceptualization as a platform.

## Lessons learned and implications for future research

In this paper we explain how we have applied Aaker's framework of brand equity in a cross-national context. In keeping with previous research, this application suggested some issues with the discriminant validity of consumer-based brand equity's dimensionality (as per Aaker). In this section we reflect on our findings and provide suggestions to help researchers improve the measurement of brand equity, particularly in an international environment. The section concludes with the limitations of the study and some ideas for further research.

A possible explanation for the issue with discriminant validity may be that the dimensions we are trying to capture are conceptually close, and creating measures that discriminate will always prove problematic. This challenge is further exacerbated in cross-national settings where translation and culture-specific brand meanings and associations create additional concerns for international researchers seeking to measure consumer-based brand equity. Furthermore, respondents may find it hard to discriminate their responses when the questions are using relatively abstract expressions (e.g. "The brand has strong associations"). What an association is could be a complex, brand specific as well as individual- and culture-specific concept. This kind of abstract questions might be great as entry-level questions for qualitative researchers – but it is difficult, on reflection, to see how in a questionnaire of a quantitative study they can tap potentially a unique concept (in this example brand associations) that is different, from other concepts (in this example, say, brand awareness). These seem to be causally linked to the extent to which a person has encountered the brand or experienced it.

Researchers often model consumer-based brand equity as a higher-order factor comprising of brand awareness, brand associations, perceived quality and brand loyalty as underlying dimensions. Whilst treating this, as such, would solve the discriminant validity problems it would not solve the associated conceptual problem and can raise questions. Can researchers argue with confidence that the higher-order construct is indeed brand equity? And, to what extent is it meaningful to aggregate individual dimension scores into a composite brand equity score, rather than keep them separate as distinct facets of the brand asset? Researchers are encouraged to revisit the conceptualization of brand equity as a construct and consider alternative conceptualizations including that of brand equity as a process which if successful, would result in certain benefits for firms and consumers. Consumer-based brand equity, in particular, is about responses to brand knowledge (Keller, 1993), so a view of this as simply a set of factors ignores the dynamic nature of its components. Dimensions of brand equity (e.g. awareness or associations) are expected to lead to changes in brand equity outcomes (e.g. attitudes or purchase intentions, see Yoo and Donthu, 2001), and it is, in fact, this change (the response – the relative change in attitudes or purchase intentions) that represents consumer-based brand equity rather than the absolute value of dimensions or outcome variables). As a result, it might be best to invest research effort into trying to understand the brand equity process including the consumer-based brand equity variables that bring about a consumer response, and under what conditions the process is observable. Critically, of course, in cross-national settings, the processes by which consumer-based brand equity facets (such as images features and so on) bring about brand equity outcomes (such as differential purchase intentions or willingness to pay a price premium) may differ across countries. Therefore, theory testing needs to be undertaken with a view to identify and explain potential differences.

Additionally, consumer-based brand equity dimensions that bring about desirable outcomes may vary from context to context (Christodoulides and de Chernatony, 2010). For instance, "image" will comprise certain associations in a non-profit situation (e.g. there may be ethical image facets – see, Michel and Rieunier, 2012) but may have different image facets in a luxury marketing context (where image features such as of exclusivity or hedonism may be important – see, Vigneron and Johnson, 2004). Yet in different countries, the extent to which an image of hedonism results in changes of outcome variables (such as purchase intentions) may differ – perhaps because of cultural norms and pressures (e.g. individualism levels or religiosity may impact the extent to which an image facet can be considered an "asset"). Likewise, image facets may be more or less salient across national contexts. For example, Stella Artois may not be seen as a premium beer in France but may be perceived as such in the UK.

This potential incongruity raises a number of questions, which can stimulate future research, including how image and associations come about in different countries and how they might be influenced by various factors such as country of origin effects and personal traits such as consumer ethnocentrism and world mindedness. Also, the extent to which global brands enjoy consistent consumer-based brand equity dimensions across national contexts is uncertain. If some of the brand equity dimensions are, indeed, inconsistent across countries (brand image/associations may be thought about as a dimension lacking conceptual equivalence) then measurement invariance may not be so relevant in that context. On the other hand, if we are to develop measures that are of practical and theoretical value allowing meaningful comparisons amongst countries then high levels of invariance is required. Future research is invited to look into this tension and advise researchers on the best course of action. Researchers are also encouraged to test theories across a large number of countries to test stability and to identify boundary conditions.

The findings also suggest that in the context of services and/or internet where the element of intangibility is prevalent, consumers may find it even harder to discriminate between the various dimensions of consumer-based brand equity. This may be due to brand knowledge *per se* being more abstract in this context where the lack of a tangible product around which associations can anchor may drive consumers to formulate an overall attitude (e.g. towards the brand) (see "halo" theory), but be unable to deconstruct this into individual dimensions. Researchers as well as practitioners are encouraged to move away from "off-the-rack" measures of consumer-based brand equity and consider developing their own measures on the basis of their specific industry's competitive environment as well as the brand's unique positioning.

This research is subject to limitations that, in turn, may stimulate further research. Given the concerns expressed in this study over the stability of Aaker's conceptualization of consumer-based brand equity across European countries and product categories, our findings should be verified in other, e.g. more diverse contexts. Also, rather than adopting an ad hoc approach to measuring their consumer-based brand equity, many firms monitor their brand assets over time by means of tracking studies. This allows them to identify fluctuations in their brand equity levels and take corrective action, if needed. It would, thus, be useful to observe the stability of Aaker's conceptualization over time particularly since the time we collected the data was affected by the European debt crisis. Finally, given the limitations with Aaker's conceptualization of consumer-based brand equity, researchers are encouraged to revisit the concept and its dimensionality and to develop and validate new measures using multi-country data.

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## Appendix

| | |
|---|---|
| *Brand awareness* | |
| I am generally aware of this brand | Lehmann *et al.* (2008), Yoo and Donthu (2001) |
| I am aware of this brand | Bravo *et al.* (2007), Im *et al.* (2012), Lehmann *et al.* (2008), Washburn and Plank (2002), Yoo and Donthu (2001), Yoo *et al.* (2000) |
| I am quite familiar with this brand[a] | Lehmann *et al.* (2008), Shah (2012), Tong and Hawley (2009) |
| I have heard of this brand[a] | Lehmann *et al.* (2008) |
| Most people are aware of this brand | Lehmann *et al.* (2008) |
| I can recognize this brand among other brands[a] | Atilgan *et al.* (2005), Bravo *et al.* (2007), Im *et al.* (2012), Shah (2012), Tong and Hawley (2009), Washburn and Plank (2002) |
| *Brand associations* | |
| Some characteristics of this brand come to my mind quickly | Bravo *et al.* (2007), Washburn and Plank (2002), Yoo *et al.* (2000) |
| I have a difficulty imagining this brand in my mind (r) | Yoo and Donthu (2001), Yoo *et al.* (2000) |
| This brand has strong associations[a] | Coleman *et al.* (2011), Keller (1993) |
| This brand has unique associations | Tong and Hawley (2009), Shah (2012) |
| This brand has favourable associations[a] | Coleman *et al.* (2011), Keller (1993) |
| I can quickly recall the symbol, logo or colour of this brand | Kim and Hyun (2011), Yoo and Donthu (2001), Yoo *et al.* (2000), |
| It is clear what this brand stands for[a] | Aaker (1991) |
| *Perceived quality* | |
| This brand is good quality[a] | Atilgan *et al.* (2005), Im *et al.* (2012), Kim and Hyun (2011), Lehmann *et al.* (2008), Pappu *et al.* (2005), Washburn and Plank, 2002; |
| This brand is consistent quality | Kim and Hyun (2011), Pappu *et al.* (2005), Tong and Hawley (2009) |
| This brand has excellent features[a] | Pappu *et al.* (2005), Shah (2012) |
| Compared to other brands in its category, this brand is of very high quality[a] | Atilgan *et al.* (2005), Tong and Hawley (2009) |
| This brand is the best brand its category | Netemeyer *et al.* (2004) |
| This brand consistently performs better than all the other brands in its category | Kim and Hyun (2011), Netemeyer *et al.* (2004) |
| I can always count on this brand for consistent high quality | Lehmann *et al.* (2008), Netemeyer *et al.* (2004) |
| This brand performs well | Lehmann *et al.* (2008) |
| This brand is effective | Lehmann *et al.* (2008) |
| This brand lives up to its promises | Lehmann *et al.* (2008) |
| This brand has served me well | Lehmann *et al.* (2008) |
| *Brand loyalty* | |
| I feel loyal to this brand [a] | Bravo *et al.* (2007), Im *et al.* (2012), Pappu *et al.* (2005), Yoo and Donthu (2001) |
| This brand is my first choice[a] | Im *et al.* (2012), Pappu *et al.* (2005), Shah (2012), Tong and Hawley (2009), Washburn and Plank (2002), Yoo and Donthu (2001), Yoo *et al.* (2000) |
| I would be willing to pay a higher price for this brand over other brands | Im *et al.* (2012), Lehmann *et al.* (2008), Shah (2012), Sung and Kim (2010), Tong and Hawley (2009), Yasin *et al.* (2007) |

(*continued*)

**Table AI.**
Item pool

| | |
|---|---|
| I intend to keep purchasing/using this brand | Sung and Kim (2010) |
| I am committed to this brand[a] | Sung and Kim (2010) |
| I will buy/use this brand the next time I need this product/service | Shah (2012), Sung and Kim (2010), Yasin *et al.* (2007) |
| I consider myself loyal to this brand | Bravo *et al.* (2007), Lehmann *et al.* (2008), Yoo and Donthu (2001), Yoo *et al.* (2000) |
| I will not buy/use other brands if this brand is not available | Washburn and Plank (2002), Yasin *et al.* (2007), Yoo and Donthu (2001) |
| If this brand were not available, it would make little difference to me if I had to choose another brand | Raju *et al.* (2009) |
| I can see myself as being loyal to this brand | Raju *et al.* (2009) |
| I will more likely purchase a brand that is on sale than this brand (r) | Raju *et al.* (2009) |

**Table AI.**

**Note:** [a]Item included in the final scale



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